

A Game-Theoretic Analysis of Loot Box Revenue Optimization: Multi-Tier Rewards and Player Heterogeneity

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Abstract

Loot boxes represent a significant and controversial component of the modern gaming industry, generating billions in revenue annually. This paper develops a comprehensive game-theoretic model to determine what reward probabilities maximize developer revenue while maintaining sustainable player engagement. Using a Stackelberg framework with heterogeneous player types and multi-tier reward structures, I implement Monte Carlo simulation with 1,000 iterations to capture realistic variance in player behavior. My analysis reveals several key findings. First, medium-rarity items with approximately 4% drop rates generate 38% of total revenue, demonstrating an optimal accessibility-desirability trade-off. Second, I discover a "Pity Paradox", guaranteed reward systems decrease revenue by 14% in multi-tier contexts, contrary to conventional wisdom. Third, player heterogeneity creates approximately 11% natural variance in revenue predictions, making distributional forecasts essential. My model predictions align closely with successful industry practice: my 4% optimal rate matches CS:GO's 3.2% Classified tier and Genshin Impact's 5.1% 4-star drop rate. The model provides both theoretical insights into optimal mechanism design and practical guidance for the \$180+ billion gaming industry.

1 Introduction

Loot boxes, randomized digital rewards purchasable and earnable in video games, have become one of the most profitable yet controversial mechanics in the gaming industry. The first video game to implement loot boxes was ZT Online in 2006, and this feature quickly became popular among developers worldwide. While some games allow players to obtain the same items through regular gameplay, the time investment required to do so is often substantially greater than purchasing loot boxes directly [1]. Developers merge elements of gambling and behavioral economics, encouraging players to continue purchasing until they finally obtain their desirable items. [2]

The modern gaming industry generated over \$180 billion in revenue in 2024, with loot boxes and similar randomized reward mechanisms representing a significant fraction of this total through free-to-play monetization strategies. [3] [4] Two of the most prominent contemporary examples

are Genshin Impact (2020) and Honkai: Star Rail (2023), both developed by HoYoverse. As of 2024, these titles continue to dominate global revenue charts through meticulously designed gacha mechanics, which share fundamental similarities with traditional loot box systems [5]. By its second anniversary, Genshin Impact had generated about \$3.7 billion USD in global player spending on mobile alone, illustrating the enormous commercial potential of gacha-style randomized reward systems in free-to-play games. [6]

Developers face a critical strategic question when designing these systems: what reward probability maximizes revenue while maintaining sustainable player engagement? This question is more complex than it initially appears. A generous drop rate for high-tier items may seem player-friendly, but it can actually reduce profit per box and thus decrease overall engagement by eliminating the scarcity value that makes rare items desirable. Players may obtain their desired items quickly and then stop purchasing boxes entirely. Conversely, excessively low drop rates frustrate players, causing them to perceive the system as unfair and disengage from the game entirely, hence potentially leading to negative reviews and a loss in player base.

This question matters for several reasons beyond pure profitability. First, the reward schedules used in loot boxes activate the same neural reinforcement systems as gambling, raising concerns about player welfare and addiction potential [7]. Understanding optimal design can inform both developer practice and regulatory policy. Secondly, formal models of loot box pricing and design treat them as probabilistic selling mechanisms, and the resulting insights on screening, pricing, and uncertainty carry over directly to mechanism design questions in subscription and freemium business models. [8]

1.1 Previous Work and my Contribution

Prior research on loot boxes has examined the phenomenon primarily from psychological, sociological, and regulatory perspectives. King and Delfabbro (2014) demonstrated clear cognitive similarities between loot box mechanics and problem gambling behaviors, showing that both activate similar reward pathways in the brain [9]. Drummond and Sauer (2018) provided empirical evidence through controlled experiments that loot boxes are "psychologically akin to gambling," finding that participants who engaged with loot box mechanics showed similar risk-taking behaviors to those observed in traditional gambling contexts [10]. However, these studies focus primarily on identifying potential harms rather than deriving optimal design principles from a game-theoretic perspective.

From an economic standpoint, existing models of digital goods and randomized rewards typically make simplifying assumptions that limit their applicability to real-world loot box systems. Most prior work assumes homogeneous consumer populations and single-tier reward structures. These assumptions dramatically oversimplify the strategic landscape that developers actually face, where player populations are highly heterogeneous in their willingness to pay and where games typically offer multiple tiers of rewards with different rarity levels.

My model extends the existing literature in three substantive ways. First, I implement a multi-

tier reward structure featuring four distinct tiers (Common, Rare, Epic, and Legendary) with varying drop rates and player-specific valuations. This reflects actual industry practice, as virtually all successful loot box games employ tiered systems rather than simple binary outcomes. Second, I incorporate player heterogeneity through three distinct player types: Whales, Casuals, and Free-to-Play (F2P) users. Each is characterized by different valuations and behavioral parameters. Critically, I allow for multi-type variance through Monte Carlo simulation, recognizing that not all players behave identically. Thirdly, providing the first systematic analysis of the "pity system", which is a guaranteed reward mechanism that has become increasingly common in response to regulatory pressure and player demands for fairness, preventing players from getting extreme bad luck.

2 Model

2.1 Game Structure and Stackelberg Framework

I model the loot box design problem as a Stackelberg game, a sequential-move game where one player (the leader) makes a decision first, and other players (the followers) observe this decision before making their own choices. In my context, the game developer acts as the Stackelberg leader who first determines and commits to the loot box drop rates, denoted by the vector $r = (r_{\text{Common}}, r_{\text{Rare}}, r_{\text{Epic}}, r_{\text{Legendary}})$, where each $r_{\text{tier}} \in (0, 1)$ represents the probability that a loot box contains an item of that particular tier.

Players act as Stackelberg followers who observe the announced drop rates and then decide how many loot boxes to open, denoted $x \geq 0$. This sequential structure accurately captures the reality of loot box markets: developers must commit to and publicly disclose drop rates (often mandated by gambling regulations in jurisdictions like China, Japan, and several European countries), and players make purchasing decisions given those rates. The developer cannot easily change rates after launch without significant reputation costs and potential regulatory violations.

The solution concept is a subgame perfect equilibrium (SPE), obtained through backward induction. I first solve for players' optimal responses to any given drop rate vector, then solve for the developer's optimal choice of drop rates anticipating those responses. This approach ensures that all strategies are optimal at every decision node, not just on the equilibrium path.

2.2 Players: Types, Preferences, and Behavior

The player population is heterogeneous, consisting of three distinct types that differ in their willingness to pay and behavioral characteristics. This typology is grounded in industry data on spending distributions in free-to-play games, where a small fraction of players generates the majority of revenue.

2.2.1 Player Type Distribution

The population consists of:

- **Whales** ($\alpha_W = 5\%$ of population): High-spending players willing to invest hundreds or thousands of dollars to obtain rare items. These players value prestige and competitive advantage.
- **Casuals** ($\alpha_C = 30\%$ of population): Moderate spenders who occasionally purchase loot boxes but are more price-sensitive than whales. These players seek enjoyment and moderate collection goals.
- **Free-to-Play (F2P)** ($\alpha_F = 65\%$ of population): Players who spend minimally or not at all. These players provide ecosystem value through network effects and player base size.

These population fractions are chosen to reflect the industry patterns in free to play games, where a small minority of high spenders generates disproportionate revenue.

2.2.2 Player Utility Function

Each player type $i \in \{W, C, F\}$ chooses the number of boxes $x_i \geq 0$ to open, seeking to maximize their utility. The utility function captures three key elements: the expected value of rewards obtained, the monetary cost of purchasing boxes, and the psychological fatigue from repeatedly engaging with the loot box mechanic.

The utility function for player type i is:

$$U_i(x_i; \mathbf{r}, f_i) = \sum_{\text{tier} \in \{C, R, E, L\}} v_{i, \text{tier}} \cdot [1 - (1 - r_{\text{tier}})^{x_i}] - p \cdot x_i - f_i \cdot x_i^2 \quad (1)$$

where the components are:

- $v_{i, \text{tier}}$: Player i 's subjective valuation for obtaining at least one item of the given tier. These valuations differ across player types and across tiers.
- r_{tier} : The drop rate (probability) for that tier, chosen by the developer.
- $1 - (1 - r_{\text{tier}})^{x_i}$: The probability of obtaining at least one item of that tier when opening x_i boxes. This follows from the complement of the probability of getting zero items, which is $(1 - r_{\text{tier}})^{x_i}$.
- p : The price per loot box, set at \$3.86 based on CS:GO weapon case prices.¹
- $f_i \cdot x_i^2$: The fatigue penalty, representing the psychological cost of repeatedly opening boxes. The quadratic form captures the realistic feature that fatigue accelerates—opening the 100th box is far more tedious than opening the 10th.

¹Calculated based on <https://github.com/jonese1234/Csgo-Case-Data/blob/master/latest.json>

The fatigue term deserves special attention. Empirical studies of consumer search and information acquisition consistently find that search costs increase more than linearly with search intensity. In the loot box context, players report feeling increasingly "burned out" and "frustrated" as they open more boxes without obtaining desired items. The quadratic specification $f_i \cdot x_i^2$ captures this accelerating psychological cost, where f_i is a player-type-specific fatigue coefficient. [10]

2.2.3 Multi-Tier Reward Structure

Real-world loot box systems invariably feature multiple tiers of rewards with different rarity levels. I model f_i tiers based on a generally prescribed system, which has become an industry standard:

Tier	Drop Rate	Whale Value	Casual Value	F2P Value
Common	50%	\$30	\$15	\$8
Rare	15%	\$150	\$60	\$25
Epic	4%	\$400	\$180	\$60
Legendary	1%	\$1,000	\$300	\$100

These valuations reflect several considerations. First, they capture the opportunity cost of players' time and money, what they would be willing to pay to obtain an item with certainty rather than gambling for it. Second, they reflect the different utility that different player types derive from items. Whales value rare items much more highly because of prestige and status signaling within the game community. Third, for F2P players, values represent the shadow price of in-game currency earned through gameplay time investment.

The drop rates follow a pattern commonly observed in successful games: Common items are guaranteed (50% means players typically get one per two boxes), Rare items provide regular positive reinforcement (15%), Epic items require moderate investment (4%), and Legendary items are aspirational (1%). This structure creates a "reward ladder" that maintains engagement across different spending levels.

2.2.4 Player Heterogeneity and Monte Carlo Simulation

A critical innovation in my model is the recognition that players within each type are not identical. While all Whales are high spenders, some Whales have higher fatigue thresholds than others. Some Casual players are closer to Whales in behavior, while others lean more toward F2P patterns.

I model this within-type heterogeneity through random fatigue coefficients. Rather than assuming all Whales have exactly the same fatigue parameter f_W , sample from distributions:

- Whales: $f_W \sim \text{Uniform}(0.03, 0.08)$
- Casuals: $f_C \sim \text{Uniform}(0.10, 0.20)$
- F2P: $f_F \sim \text{Uniform}(0.40, 0.60)$

These ranges are calibrated to produce realistic engagement patterns where all player types participate (achieve positive utility) but show substantially different box-opening behavior. The uniform distribution is chosen for simplicity.

To capture the implications of this heterogeneity, I implement Monte Carlo simulation with $N = 1000$ iterations. In each iteration:

1. Sample a fatigue coefficient f for each player type from their respective distributions
2. Solve for the optimal box-opening behavior x_i^* for each player type given those fatigue values
3. Calculate developer revenue and all other quantities of interest and store the results

This produces distributional predictions rather than point estimates, allowing us to report confidence intervals on revenue, engagement, and all other model outputs.

2.2.5 Player Optimization Problem

Given drop rates $\mathbf{r}$ announced by the developer and a fatigue realization f_i , each player type solves:

$$\max_{x_i \geq 0} U_i(x_i; \mathbf{r}, f_i) \quad (2)$$

The first-order condition for an interior optimum is:

$$\sum_{\text{tier}} v_{i,\text{tier}} \cdot r_{\text{tier}} \cdot (1 - r_{\text{tier}})^{x_i - 1} = p + 2f_i x_i \quad (3)$$

The left side represents the marginal benefit of opening one additional box (sum across tiers of value times marginal probability), while the right side represents marginal cost (box price plus marginal fatigue $2f_i x_i$).

2.3 Developer: Revenue Maximization

The developer acts as the Stackelberg leader, choosing drop rates $\mathbf{r}$ to maximize expected revenue while anticipating player responses through backward induction. The developer's objective function accounts for both gross revenue from box sales and the costs of implementing and maintaining the reward system.

2.3.1 Developer Objective Function

The developer's profit function is:

$$\Pi(\mathbf{r}) = \sum_{i \in \{W, C, F\}} \alpha_i \cdot p \cdot \mathbb{E}[x_i^*(\mathbf{r}, f_i)] - \sum_{\text{tier}} c \cdot r_{\text{tier}}^2 \quad (4)$$

where:

- α_i is the population fraction of player type i
- $x_i^*(\mathbf{r}, f_i)$ is the optimal number of boxes opened by player type i given drop rates $\mathbf{r}$ and fatigue realization f_i
- $\mathbb{E}[\cdot]$ denotes expectation over the distribution of fatigue parameters
- $c = 10$ is a development cost parameter
- $c \cdot r_{\text{tier}}^2$ represents the increasing marginal cost of higher drop rates

The revenue term $\sum_i \alpha_i \cdot p \cdot \mathbb{E}[x_i^*]$ is straightforward: it's the weighted sum of expected box purchases across all player types, multiplied by the price per box. The expectation is taken over fatigue distributions because the developer doesn't know each individual player's exact fatigue level when setting rates.

The cost term $\sum_{\text{tier}} c \cdot r_{\text{tier}}^2$ deserves more discussion. Higher drop rates impose several costs on developers. First, they require better quality or more valuable items to maintain perceived fairness; for instance, if Epic items drop frequently, they must actually be impressive to justify their rarity tier. Secondly, higher drop rates may create game balance issues requiring more development effort to maintain. The quadratic form captures increasing marginal costs: moving from 1% to 2% is relatively easy, but moving from 10% to 20% becomes prohibitively expensive in terms of balancing and long-term sustainability.

The parameter $c = 10$ is calibrated to ensure that optimal drop rates fall in realistic ranges (1–50%) rather than degenerate corner solutions. Sensitivity analysis with $c \in [5, 20]$ shows that qualitative results are robust, though optimal rates shift modestly.

2.3.2 Solution Method

The developer's optimization problem exhibits the characteristic Stackelberg structure. The developer must solve:

$$\max_{\mathbf{r} \in [0,1]^4} \Pi(\mathbf{r}) \tag{5}$$

subject to the constraint that for each player type i and fatigue realization f_i , $x_i^*(\mathbf{r}, f_i)$ solves the player's utility maximization problem.

I solve this through iterated simulation. For a given drop rate vector $\mathbf{r}$:

1. Run Monte Carlo simulation with $N = 1000$ iterations, sampling fatigue values and solving for player responses in each iteration
2. Calculate expected revenue $\mathbb{E}[\Pi(\mathbf{r})]$ as the average across iterations
3. Evaluate how revenue changes with small perturbations to $\mathbf{r}$

In practice, I find that the baseline drop rate structure (50%, 15%, 4%, 1%) is already quite close to the optimum, which provides useful validation of my model calibration. The 4D-dimensional

optimization over drop rates is computationally intensive, so for most of my analysis, I fix the tier structure and focus on comparative statics and extensions.

2.4 Pity System Extension

Many modern loot box games implement "pity systems"—guaranteed reward mechanisms designed to prevent extremely unlucky outcomes that could cause player frustration and potential regulatory concerns. The most common implementation guarantees that players receive a high-tier reward after reaching a specified number of boxes without obtaining one through normal probability.

I model a pity system by modifying the probability function for the Legendary tier. Under a pity threshold N (commonly $N = 90$ in games like Genshin Impact, Honkai Star Rail, etc), the probability of obtaining at least one Legendary item becomes:

$$P_{\text{Legendary}}(x_i, N) = \begin{cases} 1 - (1 - r_{\text{Legendary}})^{x_i} & \text{if } x_i < N \\ 1 & \text{if } x_i \geq N \end{cases} \quad (6)$$

This revised equation only applies to players who reach the threshold N .

Revised equation for developers:

$$\Pi(r, N) = \sum_i \alpha_i \cdot p \cdot x_i^*(r, N) - c \cdot r^2 - \underbrace{\kappa \cdot N}_{\text{pity implementation cost}} \quad (7)$$

Implementing a pity system imposes costs on the developer. These include direct costs (system implementation, customer support, database tracking) and indirect costs (potential revenue from the guaranteed items that would otherwise require more purchases). I model total pity costs as $\kappa \cdot N$ where $\kappa = 0.05$ is a cost-per-threshold-level parameter. For $N = 90$, this amounts to \$4.50 in implementation costs per player.

The pity system analysis addresses a fundamental question: do these player-friendly mechanisms actually improve developer profitability by maintaining engagement, or do they simply transfer surplus from developers to players? As we will see, the answer depends critically on the baseline drop rate structure.

3 Results

3.1 Baseline Model: No Pity System

We begin by analyzing the model without any pity guarantee, establishing baseline predictions for revenue, engagement, and the distribution of outcomes across player types and reward tiers.

3.1.1 Aggregate Revenue and Engagement

Monte Carlo simulation with 1,000 iterations yields the following aggregate results:

Metric	Mean	95% Confidence Interval
Net Revenue per Player	\$31.75	[\$28.26, \$35.24]
Development Cost per Player	\$0.06	[—, —]
Average Boxes Opened per Player	8.22	[7.76, 8.68]

Several features warrant attention. First, the margin of error of approximately \$3.49 represents about 11% of mean revenue. This substantial variance arises entirely from player heterogeneity in fatigue parameter, accounting for known population distributions and drop rates, uncertainty about individual player characteristics creates significant revenue uncertainty. This underscores the importance of distributional forecasts rather than point estimates for business planning.

Second, development costs are quite modest relative to revenue (\$0.06 vs \$31.75), reflecting the assumption that reward system implementation has mostly fixed costs with small marginal costs from drop rate adjustments. Sensitivity analysis with $c \in [5, 20]$ shows qualitatively similar results, though optimal drop rates shift modestly with different cost structures.

Third, average engagement of 8.22 boxes per player falls comfortably within industry norms of 5–15 boxes for typical loot box events, validating my calibration. This level of engagement is sustainable, implying players aren’t burning out from excessive grinding, but they’re sufficiently engaged to generate meaningful revenue.

3.1.2 Player-Type Decomposition

Breaking down results by player type reveals the distribution of revenue and utility across the population:

Type	Boxes	Utility (\$)	Revenue/Player (\$)	Total Rev. (\$)
Whale	46.2 ± 4.9	596.90	178.30	8.92
Casual	13.7 ± 1.2	103.79	53.03	15.91
F2P	2.8 ± 0.2	10.56	10.63	6.92
ALL	8.2 ± 0.5	—	—	31.75

The decomposition reveals several important patterns. First, all player types achieve strictly positive utility, confirming that the system is individually rational, which means every type of player prefers to participate rather than staying out entirely. This is essential for sustainability. On the other hand, a system that yields negative utility for any large segment would collapse as those players exit.

Secondly, Whales open dramatically more boxes (46.2 on average) than Casuals (13.7) or F2P players (2.8), reflecting their low fatigue and high valuations. However, despite representing only 5% of the population, Whales contribute \$8.92 out of \$31.75 total revenue—about 28% of the total. This is substantial but not dominant.

Thirdly, and perhaps most surprising, Casuals generate the largest share of revenue at \$15.91 (50% of total). This reflects two factors: they represent a substantial population fraction (30%), and their moderate per-player spending (\$53.03) is large enough to dominate the F2P segment while being scaled up by their greater numbers compared to Whales.

Moreover, F2P players, despite very low per-player spending (\$10.63), contribute 22% of revenue due to their overwhelming population share (65%). This validates the free-to-play business model’s core insight: a large base of light spenders can generate substantial revenue in aggregate.

Fifth, the standard deviations reveal different levels of within-type variance. Whales show the highest absolute variance (± 4.9 boxes) but relatively low coefficient of variation. F2P players show low absolute variance but high proportional variance, reflecting their generally constrained behavior.

3.2 Epic Tier Dominance: The Accessibility-Desirability Trade-off

My most striking finding concerns the distribution of revenue across the four reward tiers. Despite the intuition that the rarest items should generate the most revenue, I find something quite different:

Tier	Drop Rate	Revenue/Player (\$)	Revenue Share (%)	Rank
Common	50%	3.47	9.0%	4
Rare	15%	9.80	25.3%	2
Epic	4%	14.73	38.1%	1
Legendary	1%	10.64	27.5%	3

Epic items, with only a 4% drop rate, generate 38.1% of total revenue. It is more than any other single tier and more than Common and Rare combined. This is my paper’s central empirical finding, and it requires careful explanation.

3.2.1 Why Epic Dominates: Mechanism Analysis

Epic tier’s dominance reflects why I termed the *accessibility-desirability trade-off*. To understand this, consider the probability that a player obtains at least one Epic item as a function of boxes opened:

$$P(\text{Epic}) = 1 - (0.96)^x$$

At the average engagement level of $x = 8.2$ boxes:

$$P(\text{Epic}) = 1 - (0.96)^{8.2} = 0.281 \approx 28\%$$

This 28% probability is in a critical sweet spot: high enough that players perceive Epic items as achievable goals worth pursuing, but low enough that Epic items maintain scarcity value and prestige. Compare this to:

- **Common** (50%): With 8.2 boxes, $P(\text{Common}) = 1 - 0.5^{8.2} \approx 99.6\%$. Common items are virtually guaranteed, making them unexciting and low-value.
- **Rare** (15%): $P(\text{Rare}) = 1 - 0.85^{8.2} = 73\%$. Most players get Rare items, providing regular positive reinforcement, but they’re common enough to have limited value.
- **Legendary** (1%): $P(\text{Legendary}) = 1 - 0.99^{8.2} = 7.9\%$. Only about 8% of players at average engagement obtain Legendary items. For most players, Legendary items feel unreachable, limiting their impact on purchasing decisions.

Epic items sit at the optimum: the 28% probability means roughly one in four players will obtain an Epic item through typical engagement. This is low enough that obtaining one feels like an achievement, but high enough that it seems worth trying for. Players who don’t get an Epic item in their first 8 boxes often think, "I’m close—just a few more tries" rather than "This is hopeless."

3.2.2 Player-Type Contributions to Epic Revenue

Decomposing Epic tier revenue by player type reveals interesting heterogeneity:

Player Type	Epic Probability	Epic Value	Contribution
Whale	$1 - 0.96^{46.2} = 85.7\%$	\$400	\$17.14
Casual	$1 - 0.96^{13.7} = 41.5\%$	\$180	\$22.41
F2P	$1 - 0.96^{2.8} = 10.8\%$	\$60	\$4.21

Whales have an 85.7% probability of obtaining Epic items given their high box-opening rate, so they value this tier very highly. However, Casuals contribute the most to Epic revenue (\$22.41) because they combine a meaningful probability (41.5%) with both substantial value (\$180) and large population (30%).

This pattern shows how the medium-engagement segment drives the most revenue; It differs from naive intuitions that either high-spenders or the broad base should dominate. It reflects the core economic insight: revenue is the product of probability * value * population, and Epic tier optimizes all three simultaneously.

3.2.3 Comparison to Alternatives

Why doesn’t Legendary tier dominate, given its higher per-item value? The problem is accessibility. While Legendary items are valued at \$300–\$1,000 depending on player type, the 1% drop rate means most players see such low acquisition probability that it barely factors into their box-opening decisions. Whales do pursue Legendary items aggressively (7.9% probability isn’t negligible for them), which explains Legendary’s respectable 27.5% revenue share. But Legendary can’t match Epic’s broader appeal.

Similarly, Common and Rare items fail to dominate because, while accessible, they lack sufficient scarcity value. Common items are essentially free goods at typical engagement levels—nearly

everyone gets multiple copies. Rare items provide some value through moderate scarcity (27% don't get them), but not enough to overcome their lower per-item value.

Epic tier thus achieves a local optimum in the accessibility-desirability space. Moving to higher drop rates (5–10%) would increase accessibility but destroy scarcity value more than proportionally. Moving to lower rates (2–3%) would increase scarcity value but make items feel unattainable for too many players.

3.3 Player Heterogeneity Effects

The Monte Carlo approach allows us to quantify the importance of within-type variance. Consider Whale behavior. With fatigue uniformly distributed on $[0.03, 0.08]$:

- High-fatigue Whales ($f \approx 0.08$): open approximately 38 boxes
- Low-fatigue Whales ($f \approx 0.03$): open approximately 56 boxes
- **Range: $\pm 47\%$ around mean of 46.2 boxes**

This substantial within-type variance has important implications. First, it demonstrates that assuming all Whales behave identically would significantly underestimate outcome variance—by approximately 40% in my calibration. Second, it suggests that player segmentation based purely on past behavior may be unreliable; two Whales with different fatigue parameters may exhibit very different trajectories despite similar underlying preferences. Third, it means that revenue variance from heterogeneity ($\pm 11\%$) is comparable to variance from other sources of uncertainty (market conditions, seasonal effects), making it essential for risk management.

3.4 Pity System Analysis: The Pity Paradox

We now examine how guaranteed reward mechanisms affect profitability. The analysis compares the baseline (no pity) system to a pity system guaranteeing a Legendary item after $N = 90$ boxes without one, matching Genshin Impact and Honkai Star rail's design.

3.4.1 Comparative Results

System	Net Revenue	Engagement	Change
No Pity	\$31.75 $\pm$ \$3.49	8.22 $\pm$ 0.46	baseline
Pity ($N = 90$)	\$27.28 $\pm$ \$3.34	8.23 $\pm$ 0.44	−\$4.47 (−14.1%)

The pity system decreases net revenue by \$4.47 per player, a 14.1% reduction. Crucially, player engagement remains essentially unchanged (8.23 vs 8.22 boxes). This means the pity system represents pure cost without any offsetting engagement benefit, a finding we call the **Pity Paradox**.

3.4.2 Mechanism: Why Pity Reduces Profitability

The Pity Paradox arises from the interaction between multi-tier rewards and guarantee mechanisms. Several factors contribute:

1. Multi-tier structure already provides frequent rewards. With Common items at 50%, Rare at 15%, and Epic at 4%, players experience regular wins without needing guarantees. The probability of obtaining *at least one* Epic or better item in 8 boxes is approximately 30%, providing positive reinforcement frequently enough to maintain engagement.

2. Very few players approach the pity threshold. In my simulations, only 1.2% of players open 90 or more boxes. For the other 98.8%, the pity guarantee is irrelevant—they never reach it. This means the guarantee affects almost no one’s behavior, providing no engagement boost.

3. Implementation costs are significant. The pity system costs $\kappa \cdot N = 0.05 \times 90 = \4.50 per player in my calibration. This includes system implementation, database tracking, customer support, and foregone revenue from guaranteed items. Given that only 1.2% of players benefit, the per-beneficiary cost is approximately \$375—far exceeding any plausible value to those players.

4. Reduced scarcity value. The guarantee may paradoxically reduce players’ willingness to pay by making Legendary items less scarce. If “anyone who grinds 90 boxes gets one,” the prestige value diminishes.

3.4.3 When Would Pity Be Valuable?

my finding that pity reduces revenue appears to contradict the widespread adoption of pity systems in successful games like Genshin Impact. This apparent contradiction reveals an important nuance: pity systems are valuable when base drop rates are very low ($< 1\%$) and players would otherwise have a negligible probability of obtaining rare items. Genshin’s 5-star characters have a 0.6% base rate. Without pity, even a dedicated player opening 50 boxes has only $1 - 0.994^{50} = 26\%$ probability of obtaining one. With pity at 90, this becomes a guarantee. This transforms player perception from “I’ll probably never get a 5-star” to “If I save enough, I’m guaranteed one,” maintaining engagement.

In contrast, my Legendary tier has a 1% base rate. At typical engagement (8 boxes), players have 7.9% probability—low, but not hopeless. Combined with Epic items providing substantial value at 4% (28% probability), players maintain hope without guarantees. The multi-tier structure itself serves a similar function to pity systems by ensuring regular positive outcomes. The key insight: **pity systems substitute for, rather than complement, multi-tier reward structures.** Games with very low base rates and few intermediate tiers benefit from pity. Games with moderate base rates and rich tier structures find pity redundant.

3.5 Real-World Validation

A critical test of any theoretical model is whether its predictions align with observed industry practice. I compared my model’s predictions to successful games:

Sice	Medium-Rarity Drop Rate	Category
my Model (Epic tier)	4.0%	Theoretical prediction
CS:GO (Classified tier)	3.2%	Industry practice
Genshin Impact (4-star)	5.1%	Industry practice

The close alignment is striking: my theoretically derived optimal Epic rate (4%) falls between CS:GO’s disclosed Classified rate (3.2%) and Genshin’s 4-star rate (5.1%). Both comparison games are extraordinarily successful—CS:GO has sustained a thriving economy for over a decade, while Genshin Impact consistently ranks among the highest-grossing mobile games globally since its 2020 launch.

This validates several aspects of my model. First, it suggests the behavioral assumptions (utility function, fatigue structure) capture key features of actual player decision-making. Second, it indicates that successful developers have converged on similar solutions to the optimization problem, whether through explicit modeling or trial-and-error learning. Third, it confirms that medium-rarity items (3–5% range) are indeed the sweet spot for monetization, precisely as the accessibility-desirability trade-off predicts.

The small differences (3.2% vs 4.0% vs 5.1%) likely reflect minor calibration variations across games. CS:GO’s slightly lower rate (3.2%) may reflect higher per-item values in its market economy. Genshin’s slightly higher rate (5.1%) for 4-stars makes sense given its even rarer 5-star tier (0.6%) providing additional aspiration.

3.6 Limitations and Improvements

Static framework: I model a one-shot interaction, excluding dynamic elements present in real games: repeated purchases over time, seasonal events, content updates, and player learning. Multi-period extensions could capture how engagement evolves as players learn drop rates and complete their collections.

Perfect information: Players know true drop rates from the outset. In reality, players may initially misperceive rates and update beliefs through experience. Bayesian learning extensions could model this process, potentially explaining why pity systems are more valuable early in a game’s lifecycle when uncertainty is highest.

No strategic interaction: Players make independent decisions without social influences. Real games feature network effects (friends’ collections matter), status signaling (showing off rare items), and peer pressure. These mechanisms could amplify rare item value, potentially shifting the optimal point toward lower drop rates.

No budget constraints: Players optimize freely without hard budget limits. Liquidity-constrained players might behave differently, though this likely binds primarily for F2P players who already show minimal spending in my model.

Calibration uncertainty: Behavioral parameters (valuations, fatigue coefficients) are not directly observed. While my calibration produces realistic patterns and matches industry practice,

alternative specifications could potentially fit the same aggregate data. Experimental or survey data on player preferences would strengthen calibration.

3.7 Extensions and Future Research

Several promising extensions could enhance my framework:

Bayesian learning dynamics. Model players who initially don't know drop rates and learn through experience. This could explain why pity systems are more valuable early (reducing uncertainty) and why disclosure requirements affect behavior. It would also capture how player sophistication evolves over time.

Dynamic pricing and box types. Allow developers to offer multiple box types at different prices with different rate structures. Premium boxes might offer higher Epic rates at higher prices. Optimal pricing schedules could be derived using second-degree price discrimination principles.

Behavioral biases models. Incorporate gambler's fallacy ("I'm due for a win"), sunk cost effects ("I've come this far"), loss aversion (players weigh losses more than gains), and hot-hand fallacy (streaks feel meaningful). These biases could substantially affect engagement and explain phenomena my rational model misses.

Secondary markets and trading. Many games allow item trading. Resale values feed back into willingness to pay for loot boxes—a rare item with high resale value is more valuable to purchase. Market equilibrium models could endogenize both primary (loot box) and secondary (trading) markets simultaneously.

Network effects and social dynamics. Social influences amplify rare item value through status signaling and peer comparison. Agent-based models could capture how network structure affects spending cascades and whether tipping points exist where loot box systems suddenly become viral or collapse.

4 Conclusion

This paper develops a comprehensive game-theoretic model of loot box design incorporating multi-tier reward structures and player heterogeneity. Using a Stackelberg framework where developers choose drop rates and players optimize box-opening behavior, I implement Monte Carlo simulation with 1,000 iterations to capture realistic variance in outcomes.

Firstly, the accessibility-desirability trade-off. Medium-rarity items with approximately 4% drop rates generate 38% of total revenue, more than any other single tier. By optimally balancing achievability (28% probability with typical engagement) and scarcity value. This insight formalizes the intuition that items must be rare enough to be desirable but common enough to feel attainable.

Secondly, the Pity Paradox. Guaranteed reward systems decrease revenue by 14% in multi-tier contexts, contrary to conventional wisdom. This occurs because multi-tier structures already provide adequate engagement through frequent rewards at Common (50%), Rare (15%), and Epic

(4%) tiers. Pity guarantees impose implementation costs (\$4.50) while affecting very few players (1.2% reach thresholds), yielding no net engagement benefit. The key insight: pity systems substitute for rather than complement multi-tier structures.

Thirdly, real-world validation. my model’s 4% optimal Epic rate aligns closely with disclosed rates in extraordinarily successful games: CS:GO’s 3.2% Classified tier and Genshin Impact’s 5.1% 4-star rate. This tight correspondence across independently designed systems validates behavioral assumptions and suggests the model captures fundamental mechanisms.

The answer to my central research question is clear: **a multi-tier reward structure with Epic items at approximately 4% drop rate maximizes developer revenue while maintaining sustainable player engagement, generating \$31.75 ± \$3.49 per player (95% confidence interval).** This design achieves optimal balance by ensuring all player types→ Whales (5% population, 28% revenue), Casuals (30% population, 50% revenue), and F2P (65% population, 22% revenue), all participated with positive utility.

Looking forward, incorporating Bayesian learning, behavioral biases, and multi-period dynamics could further refine optimal designs while addressing ongoing ethical concerns about gambling-like mechanics in games. The tension between profitability and player welfare remains a central challenge, and my framework provides a foundation for analyzing how different design choices affect both objectives.

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